

Quantum Entanglement as Structural Non-Closure: A TNA Perspective

Abstract

Quantum entanglement violates Bell inequalities experimentally, ruling out locally closed descriptions of physical systems. This violation is not dynamical—no superluminal signaling occurs—but structural: the global quantum state cannot be reconstructed from local properties. We formalize this failure of local closure using the Theory of Axiomatic Necessity (TNA), where coherent systems require irreducible external structural support (N_1) beyond their locally observable domain (N_0). Entanglement provides a physical instantiation of this principle: the global wavefunction plays a role structurally analogous to N_1 , enabling coherence without operating causally. This framework unifies quantum non-separability with similar structural limits in mathematics, machine learning, and social systems, demonstrating that local closure is not merely difficult but impossible for coherent systems.

1. Introduction: The Intuition of Local Closure

Classical physics and everyday intuition share a premise: the world is locally self-contained. Understanding this thing here should not require looking elsewhere. Explanations can be boxed, bounded, closed.

Quantum entanglement ruins this intuition—not with drama, but with experimental precision.

Since Einstein, Podolsky, and Rosen (1935) questioned quantum completeness, and Bell (1964) derived testable inequalities, experiments have consistently violated local realism. The correlations observed between entangled particles cannot be explained by any theory assigning definite local states to each particle.

Yet the violation is subtle. No information travels faster than light. No causality is breached. Something else fails: the very possibility of local explanation.

2. The Structural Nature of the Failure

Popular accounts often frame entanglement as "spooky action at a distance"—mysterious influence between distant particles. This misses the point.

The problem is not dynamics. It is structure.

When two particles are entangled, their joint state is irreducible. No description of particle A alone, however complete, combined with any description of particle B alone, however complete, can reproduce the observed correlations. The system only makes sense as a whole.

This is not a limitation of our knowledge. It is a limitation of local ontology.

Experiments violating Bell inequalities (Aspect et al., 1982; Weihs et al., 1998; Hensen et al., 2015) demonstrate decisively: you can keep locality, or you can keep realism, but you cannot keep both. The conjunction fails not because particles communicate, but because local properties are insufficient for global coherence.

3. TNA Framework: N_0 , N_1 , and Structural Coherence

The Theory of Axiomatic Necessity (TNA) distinguishes two levels in coherent systems:

- N_0 : the domain of locally observable, measurable, describable phenomena.
- N_1 : the minimal structural support required for N_0 to remain coherent, not derivable from N_0 itself.

TNA's central result: no system producing observables can fully account for its own conditions of coherence without invoking external structural support. Closure is not merely difficult; it is structurally impossible.

Key properties of N_1 :

- **Non-operational**: it does not act causally within N_0 .
 - **Non-eliminable**: removing it destroys explanatory adequacy.
 - **Non-metaphysical**: its necessity is inferred from failure modes, not postulated.
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4. Entanglement as Physical N_1

In entangled quantum systems, the global wavefunction exhibits precisely these properties:

PROPERTY	N_1 (TNA)	GLOBAL QUANTUM STATE
Not locally observable	✓	✓ (no eigenstate for single particle)
Required for coherence	✓	✓ (correlations unexplained without it)
No causal operation	✓	✓ (no signaling, no energy transfer)
Cannot be eliminated	✓	✓ (Bell violation persists)

The global state is not a hidden force or additional object. It is a structural condition: without it, local descriptions are incomplete; with it, they cohere.

This mirrors TNA's N_1 exactly: not something beyond the system, but something beyond its local description, necessary for its intelligibility.

5. Cross-Domain Structural Parallels

The pattern—coherence requiring irreducible non-local support—appears across domains:

DOMAIN	N_0 (LOCAL)	N_1 (GLOBAL/STRUCTURAL)	FAILURE OF CLOSURE
Quantum mechanics	Single-particle properties	Entangled global state	Bell inequality violation
Mathematics	Theorems derivable within system	Axioms, consistency conditions	Gödel incompleteness
Machine learning	Trained network, local parameters	Winning ticket, spectral core	Irreversible collapse when pruned below $F_{\min}$
Social systems	Individual interactions	Shared norms, price signals	Market failure when signals disappear

In each case, local closure is attempted and fails. Coherence persists only when external structural support is acknowledged.

6. Direction of Dependence

Crucially, TNA does not rely on quantum entanglement for its validity. Nor does entanglement "prove" TNA empirically.

The relationship is asymmetric: entanglement is one independently discovered instance of a general structural constraint that TNA formalizes across domains. Quantum mechanics happens to be where this constraint was measured with lasers and statistics.

This direction matters. It prevents TNA from being dismissed as "quantum mysticism" and entanglement from being overinterpreted as metaphysical.

Both describe the same limit: **coherence cannot be locally self-sustaining**.

7. Conclusion: Closure Exposed, Not Violated

Quantum entanglement did not destroy classical intuition with drama. It did something worse: it showed—calmly, repeatedly, experimentally—that local closure was never viable.

The global quantum state is not a mystery. It is a **receipt**: proof that some truths live outside the parts we try to isolate.

For physics, this means accepting non-separability as fundamental. For epistemology, it means recognizing that explanation always presupposes structural conditions it cannot fully internalize.

The universe refuses to be locally complete. And it has experimental backing.

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